

Linux Users and Groups - Study Notes

I. Users and Groups Overview

I.1 Core Concepts

- **Purpose:** Access control and permissions management
- **User Identification:** System uses User ID (UID) internally, usernames are human-friendly
- **Group Identification:** System uses Group ID (GID) for group management
- **Home Directory:** Usually located at `/home/username`

I.2 Types of Users

- **Human Users:** Regular people using the system
- **System Users:** Daemon processes that run system services
- **Root/Superuser:** Most powerful user with access to everything

I.3 Permission Example

```
# Viewing a protected file without permissions
$ cat /etc/shadow
cat: /etc/shadow: Permission denied

# Check file permissions
$ ls -la /etc/shadow
-rw-r----- 1 root shadow 1134 Dec 1 11:45 /etc/shadow
# Breakdown: root owns file, shadow group can read, others have no access

# Access with sudo
$ sudo cat /etc/shadow
# Now displays file contents
```

Example Output:

```
root:$6$salt$hashedpassword...:18500:0:99999:7:::
daemon:*:18500:0:99999:7:::
bin:*:18500:0:99999:7:::
```

2. Root Access and Superuser Commands

2.1 sudo Command (Recommended)

```
# Run single command as root
$ sudo command_name

# Example: View system files
$ sudo cat /etc/shadow
$ sudo ls /root
$ sudo systemctl restart nginx
```

Example sudo Session:

```
# Regular user tries to access root directory
$ ls /root
ls: cannot open directory '/root': Permission denied

# Using sudo
$ sudo ls /root
Desktop  Documents  Downloads  # Shows contents
```

2.2 su Command (Switch User)

```
# Switch to root user (opens root shell)
$ su
Password: [enter root password]
# Now you're in root shell - be careful!

# Switch to specific user
$ su - username
Password: [enter user's password]

# Exit back to original user
$ exit
```

Example su Session:

```
john@computer:~$ su
Password: *****
root@computer:/home/john# whoami
root
```

```
root@computer:/home/john# exit
john@computer:~$ whoami
john
```

2.3 Sudoers File

```
# Edit sudoers file (always use visudo)
$ sudo visudo

# View current sudoers (don't edit directly)
$ sudo cat /etc/sudoers
```

Example /etc/sudoers content:

```
root    ALL=(ALL:ALL) ALL
%admin   ALL=(ALL) ALL
%sudo    ALL=(ALL:ALL) ALL
john     ALL=(ALL) NOPASSWD: /usr/bin/systemctl
```

3. /etc/passwd File

3.1 Viewing User Information

```
# View all users
$ cat /etc/passwd

# View specific user
$ grep username /etc/passwd

# Count total users
$ wc -l /etc/passwd
```

3.2 Understanding /etc/passwd Format

Field Structure: username:password:UID:GID:GECOS:home_directory:shell

Example Entry:

```
john:x:1001:1001:John Smith,Room 101,555-1234:/home/john:/bin/bash
```

Field Breakdown:

1. **Username:** john
2. **Password:** x (stored in /etc/shadow)
3. **UID:** 1001 (User ID)
4. **GID:** 1001 (Primary Group ID)
5. **GECOS:** John Smith,Room 101,555-1234 (Real name, office, phone)
6. **Home Directory:** /home/john
7. **Shell:** /bin/bash

3.3 Special User Examples

```
# System users examples from /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
```

Analysis:

- root : UID 0 (superuser), has login shell
- daemon : UID 1, system user, no login shell
- www-data : Web server user, no login access

3.4 Password Field Symbols

- x : Password stored in /etc/shadow
- * : User has no login access
- (blank) : User has no password

Example:

```
john:x:1001:1001:John Smith:/home/john:/bin/bash      # Normal user
guest*:1002:1002:Guest Account:/home/guest:/bin/bash  # No login
test::1003:1003:Test User:/home/test:/bin/bash        # No password
```

4. /etc/shadow File

4.1 Viewing Shadow File

```
# Requires sudo access
$ sudo cat /etc/shadow

# View specific user's shadow entry
$ sudo grep username /etc/shadow
```

4.2 Understanding /etc/shadow Format

Field Structure:

```
username:encrypted_password:last_change:min_age:max_age:warn_period:inactive_per-
iod:expiration_date:reserved
```

Example Entry:

```
john:$6$randomsalt$longhashpassword:18500:0:99999:7:30:19000:
```

Field Breakdown:

1. **Username:** john
2. **Encrypted Password:** \$6\$randomsalt\$longhashpassword
3. **Last Change:** 18500 (days since Jan 1, 1970)
4. **Min Age:** 0 (days before password can be changed again)
5. **Max Age:** 99999 (days before password must be changed)
6. **Warning Period:** 7 (days before expiration warning)
7. **Inactive Period:** 30 (days after expiration before account locks)
8. **Expiration Date:** 19000 (account expires on this date)
9. **Reserved:** Empty field for future use

4.3 Password Field Examples

```
# Active user with password
john:$6$salt$hash:18500:0:99999:7:::

# Locked account
mary: !$6$salt$hash:18500:0:99999:7:::

# No password set
guest: *:18500:0:99999:7:::
```

```
# Account disabled
disabled:!:18500:0:99999:7:::
```

4.4 Date Calculation Example

```
# Calculate days since Jan 1, 1970
$ date -d "1970-01-01 + 18500 days"
Mon Aug 15 00:00:00 UTC 2020

# Current date in days since epoch
$ echo $(( $(date +%s) / 86400 ))
```

5. /etc/group File

5.1 Viewing Group Information

```
# View all groups
$ cat /etc/group

# View specific group
$ grep groupname /etc/group

# View groups for current user
$ groups

# View groups for specific user
$ groups username
```

5.2 Understanding /etc/group Format

Field Structure: `group_name:password:GID:user_list`

Example Entries:

```
root:*:0:
sudo:*:27:john,mary
developers:*:1500:john,alice,bob
marketing:*:1501:mary,sarah
```

Field Breakdown:

1. **Group Name:** developers
2. **Password:** * (not used, sudo preferred)
3. **GID:** 1500 (Group ID)
4. **User List:** john,alice,bob (comma-separated)

5.3 Group Examples with Users

```
# View current user's groups
$ groups
john adm cdrom sudo dip plugdev lpadmin sambashare

# View specific user's groups
$ groups alice
alice : alice developers marketing

# View who's in sudo group
$ grep sudo /etc/group
sudo:x:27:john,mary,admin
```

5.4 Primary vs Secondary Groups

```
# Primary group (from /etc/passwd)
$ grep john /etc/passwd
john:x:1001:1001:John Smith:/home/john:/bin/bash
#           ^ Primary GID (1001)

# Secondary groups (from /etc/group)
$ grep john /etc/group
sudo:x:27:john,mary
developers:x:1500:john,alice
```

6. User Management Tools

6.1 Adding Users

useradd Command (Basic)

```
# Basic user creation
$ sudo useradd bob

# Create user with home directory
$ sudo useradd -m bob

# Create user with specific shell
$ sudo useradd -m -s /bin/bash bob

# Create user with specific UID
$ sudo useradd -m -u 1500 bob

# Create user with multiple groups
$ sudo useradd -m -G sudo,developers bob
```

adduser Command (Interactive)

```
# Interactive user creation (Debian/Ubuntu)
$ sudo adduser bob
Adding user `bob' ...
Adding new group `bob' (1002) ...
Adding new user `bob' (1002) with group `bob' ...
Creating home directory `/home/bob' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
```

6.2 Removing Users

```
# Remove user (keep home directory)
$ sudo userdel bob

# Remove user and home directory
$ sudo userdel -r bob

# Remove user and all files
$ sudo userdel -r --force bob
```

Example Deletion Process:

```
# Before deletion
$ ls /home
```



```
alice bob john

$ grep bob /etc/passwd
bob:x:1002:1002:Bob User:/home/bob:/bin/bash

# Delete user
$ sudo userdel -r bob

# After deletion
$ ls /home
alice john

$ grep bob /etc/passwd
# No output - user removed
```

6.3 Changing Passwords

```
# Change your own password
$ passwd
Changing password for john.
Current password:
New password:
Retype new password:
passwd: password updated successfully

# Change another user's password (as root)
$ sudo passwd bob
New password:
Retype new password:
passwd: password updated successfully

# Force password change on next login
$ sudo passwd -e bob
```

6.4 User Modification

```
# Change user's shell
$ sudo usermod -s /bin/zsh john

# Add user to group
$ sudo usermod -a -G sudo john

# Change user's home directory
$ sudo usermod -d /home/newjohn john
```

```
# Lock user account
$ sudo usermod -L john

# Unlock user account
$ sudo usermod -U john
```

6.5 Checking User Creation Results

```
# Check /etc/passwd entry
$ grep bob /etc/passwd
bob:x:1002:1002::/home/bob:/bin/bash

# Check /etc/shadow entry
$ sudo grep bob /etc/shadow
bob:$6$randomsalt$hashedpassword:18500:0:99999:7:::

# Check home directory
$ ls -la /home/bob
total 20
drwxr-xr-x 2 bob bob 4096 Dec 1 10:30 .
drwxr-xr-x 5 root root 4096 Dec 1 10:30 ..
-rw-r--r-- 1 bob bob 220 Dec 1 10:30 .bash_logout
-rw-r--r-- 1 bob bob 3771 Dec 1 10:30 .bashrc
-rw-r--r-- 1 bob bob 807 Dec 1 10:30 .profile
```

Quick Reference Commands

User Information

<code>whoami</code>	# Current username
<code>id</code>	# Current user ID and groups
<code>groups</code>	# Current user's groups
<code>w</code>	# Who is logged in
<code>last</code>	# Last login history

File Viewing

<code>cat /etc/passwd</code>	# All users
<code>cat /etc/group</code>	# All groups

```
sudo cat /etc/shadow # Password information
```

User Management

```
sudo useradd -m username    # Add user with home
sudo userdel -r username    # Remove user and home
sudo passwd username        # Change password
sudo usermod -a -G group user # Add user to group
```

Permission Commands

```
sudo command    # Run as root
su - username    # Switch to user
sudo visudo      # Edit sudoers file
```

Security Best Practices

1. Use **sudo** instead of **su** for administrative tasks
2. Always use **visudo** to edit `/etc/sudoers`
3. Never edit `/etc/passwd`, `/etc/shadow`, `/etc/group` directly - use proper tools
4. Use **strong passwords** and consider password aging policies
5. Remove **unused user accounts** promptly
6. Monitor **sudo usage** through logs (`/var/log/auth.log`)
7. Use **groups for permission management** rather than individual user permissions